

Roll No. 22081028021

21473

**B. Pharmacy 2nd Semester
(Regular/Re-appear/Improvement
Examination – June, 2023**

BIOCHEMISTRY THEORY

Paper : BP203T

Time : Three hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Attempt *all* questions from Section-A, *two* questions from Section-B and *seven* questions from Section-C.

SECTION – A

1. Write short note with suitable example : $2 \times 10 = 20$
- (a) Discuss biological role of Lipid.
- (b) Write significance of ATP and cyclic AMP.

P. T. O.

- (c) What is hyper bilirubinemia?
- (d) Define essential amino acids with examples.
- (e) What is Gluconeogenesis ?
- (f) Name the enzymes present in DNA replication.
- (g) Write chemical structure of following amino acids :
 - (a) Tryptophan
 - (b) Glutamic acid.
- (h) Define competitive and non-competitive enzyme inhibition.
- (i) Define Glycolysis.
- (j) Structure of DNA and RNA.

SECTION – B

Long answer questions: Answer any *two* out of the following *three* questions :

$$10 \times 2 = 20$$

2. (a) Describe urea cycle and explain the general reaction of amino acids.

- (b) Explain the factors affecting enzymes activity.
Derive Michealis-Menten equation and line weaver burke plot.
3. Write in detail about the structure and function of DNA and RNA.
4. Explain the pathway, energetics and significance of Citric acid cycle.

SECTION – C

Short answer type questions : Answer any *seven* out of the following *nine* questions : $5 \times 7 = 35$

5. Explain about HMP Pathway and its significance.
6. Describe the B-oxidation saturated fatty acids.
7. Discuss about the biological significance of Cholesterol and Vitamin D.
8. Write in detail about the catabolism of Phenylalanine and explain about electron transport chain.

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9. Explain De novo synthesis of Fatty acids.
 10. (a) Describe in detail about the Hormonal regulation of blood glucose.
(b) Describe biosynthesis of purines nucleotides.
 11. Write note on Endergonic and Exergonic reaction and discuss about the classification of energy rich compounds.
 12. Discuss the synthesis and significance of biological substance 5-HT.
 13. Write in detail about Transcription and Translation process.
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Department of Pharmaceutical Sciences,
B. Pharmacy 2nd Semester-2023 1st Sessional Examination
BP 203T Biochemistry

Time: 1.5 Hours

Marks: 30

Section-A

Attempt all questions. Each question carries 2 marks.

2x5=10

Q.1 Write a note on:

- a) Define Glycogenesis and Glycogenolysis.
- b) Define nitrogenous bases with structure.
- c) Define coenzyme with examples.
- d) Define replication with steps?
- e) How many ATP are produced in glycolysis and TCA cycle?

Section-B

Long Answer Questions: Answer anyone out of the following two questions

10x1=10

Q.2 Describe the process of DNA replication and transcription in detail.

Q.3 Give the derivation of Michaelis-Menton equation and also explain factor affecting enzyme activity.

Section-C

Short Answer Questions: Answer any two out of the following three questions

5x2=10

Q.4 Give the oxidative phase of HMP pathway. Discuss its significance.

Q.5 Describe Embden Meyerh of pathway in the body with energetic.

Q.6 Explain Gout disease. Give the structure and functions of DNA and RNA.

Department of Pharmaceutical Sciences, Indira Gandhi University, Meerpur, Rewari
B. Pharmacy 2nd Semester Second Sessional Examination
Pharmaceutical Biochemistry(Paper code- BP203T)

Duration- 1h 30 min

Max. Marks- 30

Question no.1 is compulsory. Each question carries 2 marks.

1. Answer the following.

- a. What is ketoacidosis?
- b. Write reaction of transamination and deamination.
- c. Write a note on energy rich compounds.
- d. Describe inhibitors and uncouplers of oxidative phosphorylation.
- e. Give a note on jaundice.

Attempt any two questions. Each question carries 5 marks.

2. Give a detailed note on Formation and utilization of Ketone bodies.
3. Write a note on the mechanism of oxidative phosphorylation.
4. Write a note on ETC and its inherited disorders.

Attempt any one question. Each question carries 10 marks.

5. Write a note on catabolism of heme. Detail notes on disorders of amino acid metabolism
6. Write biological significance of cholesterol and conversion of it into bile acid, vitamin D.

Indira Gandhi University, Meerpur, Rewari

Department of Pharmaceutical Sciences

B. Pharmacy 2nd Semester & LEET - 25 August 2023

Paper Code- BP 205T

COMPUTER APPLICATIONS IN PHARMACY

Time: 2 hours

Total marks: 50

Section-A

Attempt any 6 questions out of 8 questions and each question carry 5 marks.

Q1) Explain binary addition and subtraction with examples.

Q2) Explain planning and managing the project in detail.

Q3) Write note on One's complement and Two's complement method.

Q4) Define CSS and M S Access. Explain advantages and disadvantages of both.

Q5) Define DBMS. Explain examples and applications of DBMS in detail.

Q6) What do you understand by Electronic prescription? What are its benefits?

Q7) Difference between pharmacokinetics and pharmacodynamics. Explain about pharmacokinetics in detail.

Q8) i) Conversion of $(294)_{10}$ into Binary number system.

ii) Conversion of $(1010111001011100)_2$ to Hexadecimal number system.

Attempt any 2 questions out of 3 questions and each question carry 10 marks.

Q9) i) Define binary number system & its types in detail.

ii) Explain pharma information system and its benefits in detail.

Q10) Define bioinformatics. Explain objective of bioinformatics and bioinformatics databases in detail.

Q11) Explain laboratory information management system and text information management system in detail.

Indira Gandhi University, Meerpur, Rewari

Department of Pharmaceutical Sciences

B. Pharm. 2nd Semester, 1st Sessional Examination, 05 July 2023

Paper Code – BPH 205 T ; Computer applications in Pharmacy

Duration : 1 hr30 min.

Total marks

Q1 is compulsory. Each part carries 2 marks

Explain in detail

i) Patient monitoring system, ii) CSS, iii) Lab diagnostic system, iv) Barcode medicine identification, v) Databases

Attempt any two out of three. Each question carries 5 marks

Q2. What do you understand by Electronic prescription? What are its benefits?

Q3. Describe various types of programming languages in detail.

Q4. Difference between HTML and XML.

Attempt any one out of two. Each question carries 10 marks

Q5. i) Briefly describe hospital and clinical pharmacy. Also discuss applications of computers in hospital and clinical pharmacy.

ii) Write short note on Drug information storage and retrieval.

Q6. i) Explain about pharmacy drug database

ii) Write short note on MS Access

Indira Gandhi University, Meerpur, Rewari

Department of Pharmaceutical Sciences

B. Pharm. 2nd Semester, 2nd Sessional Examination, 04 August 2023

Paper Code – BPH 205 T ; Computer applications in Pharmacy

Total marks:

Duration : 1 hr30 min.

Q1 is compulsory. Each part carries 2 marks

Explain in detail

- i) Bioinformatics, ii) Nucleic acid database, iii) Any two name of bioinformatics database, iv) Proteomics,
v) Genomics

Attempt any two out of three. Each question carries 5 marks

- Q2. Explain objectives of Bioinformatics in detail.
Q3. Briefly explain the impact of bioinformatics in discovery of vaccines.
Q4. Explain Text Information Management System and its benefits.

Attempt any one out of two. Each question carries 10 marks

- Q5. i) Discuss Laboratory Information Management System (LIMS) in detail.
ii) Describe Chromatographic Data Analysis and its importance.
Q6. i) Briefly describe Bioinformatics and its objectives.
ii) Discuss classification of biological databases in detail.

Indira Gandhi University, Meerpur, Rewari

Department of Pharmaceutical Sciences

B. Pharm. 2nd Semester, Final Examination, August, 2023

Time: 2 Hrs Subject:- Environmental Sciences (Paper code – BP 206T) Maximum Marks: 50

Note: Attempt any six questions from section A and two questions from section B.

Section A (Short Answer Questions)

Q.1) Write a note on followings:

(6x5)

- a. Deforestation, its causes and consequences.
- b. Mineral Resources
- c. Grassland Ecosystem
- d. Desert Ecosystem
- e. Energy Resources
- f. Multidisciplinary nature of environmental studies.
- g. Air Pollution
- h. Natural Resources, difference between renewable and non-renewable resources.

Section B (Long Answer Questions)

(2X10)

Q.2. What you understand by ecosystem. Enumerate the aquatic ecosystem.

Q.3 Write a note on food resources and role of an individual in conservation of natural resources.

Q.4 Write a note on water and soil pollution.

B.Pharm. 2nd Sem 1st Sessional, Subject:- Environmental Sciences (Paper code – BP 206T)

Time: 1.5 Hrs

Maximum Marks: 30

Note: Section A is compulsory. Attempt any two questions from section B and one question from section C.

Section A

Q.1) Write a note on followings

(2x5)

- a. Air Pollution
- b. Mineral Resources
- c. Deforestation, its causes and consequences.
- d. Forest resources
- e. Difference between renewable and non-renewable resources.

Section B (Short Answer Questions)

Q.2 Write a note on energy resources.

(05 Marks)

Q.3 Describe the Role of an individual in conservation of natural resources.

(05 Marks)

Q.4 Write down various factors responsible for water pollution.

(05 Marks)

Section C (Long Answer Questions)

Q.5 Describe various natural resources and associated problems.

(10 Marks)

Q.6 Write a note on environmental pollution.

(10 Marks)

Department of Pharmaceutical Sciences, IGU, Meerpur
B. Pharm. 2nd Semester, 2nd Sessional Examination, August, 2023
Subject:- Environmental Sciences (Paper code – BP 206T)

Time: 1.5 Hrs

Maximum Marks: 30

Note: Section A is compulsory. Attempt any two questions from section B and one question from section C.
Section A

- Q.1) a) Define ecosystem.
b) Write down the main causes of soil pollution.
c) Define Biodiversity.
d) Differentiate between deforestation and afforestation.
e) Who coined the term ecology? Name the largest ecosystem of earth.

Section B (Short Answer Questions)

Q.2 Write a note on Desert ecosystem.

Q.3 What do you mean by Soil pollution? How do you prevent it?

(05 Marks)

Q.4 Briefly explain aquatic ecosystem.

(05 Marks)

(05 Marks)

Section C (Long Answer Questions)

Q.5 Mention different types of grassland in India. Write a short note on food chain on grassland ecosystem. (10 Marks)

Q.6 Define ecosystem. Give its classification, structure and functions in detail.

(10 Marks)

A.G. Tansley in 1953

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**B. Pharmacy 2nd Semester
(Regular/Re-appear/Improvement
Examination – June, 2023**

HUMAN ANATOMY AND PHYSIOLOGY-II THEORY

Paper : BP201T

Time : Three hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

SECTION – A

Attempt all questions. Each question carries **2** marks.

2 × 10 = 20

1. Define the following :

(a) Define meninges.

(b) Discuss Role of pepsin in protein digestion.

P. T. O.

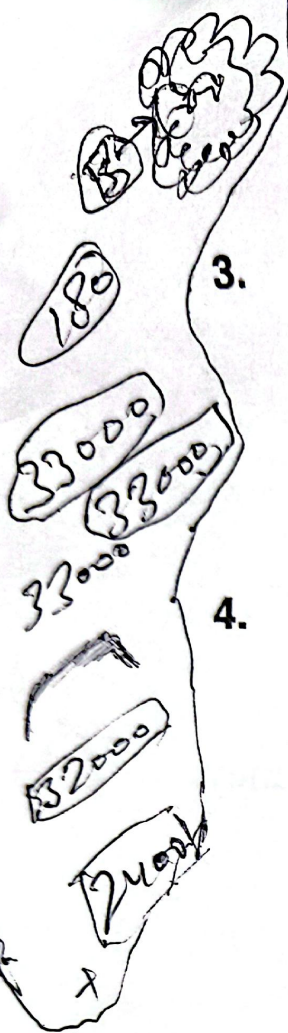
- (c) Define lung volumes.
- (d) Define classification of hormones.
- (e) Define pregnancy and parturition.
- (f) Write a note on chromosomes.
- (g) What is micturition reflex ?
- (h) Write a note on BMR.
- (i) Define neurotransmitter.
- (j) Define parathyroid gland.

SECTION – B

Long Answer Questions: Answer any *two* out of the following three questions.

10 × 2 = 20

2. Write a note on structure and functions of CNS ?
Define regulation of acid production through
parasympathetic nervous system.

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3. Write a note on structure and functions of respiratory system. Define lung volumes and discuss mechanism of respiration.
 4. Describe the anatomy and physiology of reproductive system.

SECTION - C

Short Answer Questions : Answer any *seven* out of the following nine questions.

$5 \times 7 = 35$

5. Discuss the physiology of menstruation.
6. Write a note on protein synthesis.
7. Describe the mechanism of hormone action and discuss about structure and functions of pituitary gland.
8. Write a note on physiology of urine formation.
9. What is the classification and properties of nerve fibres ?

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10. Write a short note on energetics.
 11. Discuss anatomy and physiology of GIT.
 12. Discuss in detail organisation of nervous system.
 13. Discuss role of RAS in kidney and disorder of kidney.
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Time: 1.5 Hrs

Max. Marks: 30

SECTION-A

Attempt all Questions. Each question carry 2 marks.

2X5=10

1. Artificial respiration
2. Neurotransmitter
3. Disorder of GIT
4. Lung volumes
5. Ventricles of Brain

SECTION-B

Short Answer Questions. Attempt any 2 out of Three.

5X2=10

1. Explain the various parts and function of central nervous system?
2. Define CSF? Explain various functions of Brain.
3. Explain classification and properties of nerve fibres.

SECTION-C

Long Answer type Question. Attempt any One out of Two

10X1=10

1. Explain the anatomy and physiology of digestive system with diagram
2. Discuss the mechanism of respiration in detail with different types of lung capacity.

B. Pharmacy 2nd Semester 2nd Sessional Exam August 2023
Human Anatomy and Physiology-II
Subject code: BP201T

Time: 1.5 Hrs

Max. Marks: 30

SECTION-A

Attempt all Questions. Each question carry 2 marks.

2X5=10

1. Chromosome and DNA
2. Disorders of kidney
3. Spermatogenesis
4. Disorders of thyroid gland
5. Function of adrenal gland

SECTION-B

Short Answer Questions. Attempt any 2 out of Three.

5X2=10

1. Write the structure, function and disorder of Pituitary gland.
2. Explain the classification and mechanism of hormone action.
3. Explain in detail the physiology of menstruation.

SECTION-C

Long Answer type Question. Attempt any One out of Two

10X1=10

1. Explain the physiology of urine formation with diagram.
2. Describe Anatomy and Physiology of Female or Male reproductive system.

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**B. Pharmacy 2nd Semester
(Regular/Re-appear/Improvement
Examination – June, 2023**

**PHARMACEUTICAL ORGANIC CHEMISTRY-I
THEORY**

Paper : BP202T

Time : Three hours]

] Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

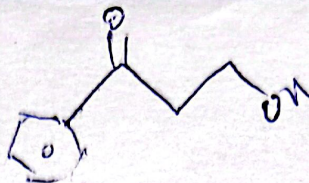
Note : Attempt *all* questions from Section-A, *two* questions from Section-B and *seven* questions from Section-C.

SECTION – A

1. Write short note with suitable example : $2 \times 10 = 20$
- (a) Stability of alkene
 - (b) Ozonolysis of Alkenes

P. T. O.

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- (c) Anti-Markonikoff's orientation
- (d) Halogenation of Alkanes
- (e) Structure of Salicylic acid
- (f) Structure and use of Cinnamaldehyde
- (g) Crossed Aldol Condensation
- (h) Cannizzaro reaction
- (i) Hybridization in Alkenes.
- (j) Qualitative test for Ethanolamine

SECTION – B

Long answer type questions : Answer any **two** out of the following **three** questions :

$$10 \times 2 = 20$$

2. Discuss the mechanism, reactivity and factors effecting in Elimination Uni-molecular reaction.
3. What are Cycloaddition reactions ? Explain Diel's alder reaction with mechanism. Explain the mechanism of Perkin's condensation.

4. Give the structure and uses of :

- (a) Benzyl alcohol
- (b) Benzaldehyde
- (c) Chloroform
- (d) Propylene glycol
- (e) Ethyl alcohol

SECTION – C

Short answer type questions : Answer any *seven* out of the following *nine* questions : $5 \times 7 = 35$

5. Write the mechanism and orientation of SN' reaction with suitable example.

6. Give the structure and uses of :

- (a) Vanillin
- (b) Benzaldehyde
- (c) Chloroform
- (d) Propylene glycol
- (e) Ethylenediamine

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7. Explain the acidity of carboxylic acid with suitable examples.
8. Explain the mechanism of Perkin's condensation.
9. Discuss the acidity of carboxylic acid. Write the statement and reaction mechanism of aldol condensation.
10. Explain Qualitative test, structure and use of Methyl Alcohol.
11. Write a note on Aldol condensation and Crossed Aldol condensation.
12. Write in detail about Markonikoff's and Anti Markonikoff's orientation.
13. Discuss about the Structure, Qualitative test, use of :
 - (a) Amphetamine
 - (b) Ethanolamine



Department of Pharmaceutical Sciences, Indira Gandhi University, Meerpur, Rewa
B. Pharmacy 2nd Semester Second Sessional Examination
Pharmaceutical Organic Chemistry-I (Paper code- BP202T)

Max

Duration- 1h 30 min

Question no.1 is compulsory. Each question carries 2 marks.

1. Answer the following.

- What is Satzefff rule? Give its reaction.
- Write reaction of Markownikoff's rule.
- Write Koch reaction for carboxylic acid.
- Give the structure and two uses of trichloroethane.
- Give peroxide effect with reaction.

Attempt any two questions. Each question carries 5 marks.

- Give a detailed note on acidity of carboxylic acid and effect of substituent's on it.
- Give the mechanism, reactivity and factors affecting E₂ reaction.
- Write four methods of preparation alcohols.

Attempt any one question. Each question carries 10 marks.

- Write physical and chemical properties of amines in detail.
- Write method of preparation and chemical properties of alkyl halides.



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**B. Pharmacy 2nd Semester
(Regular/Re-appear/Improvement)
Examination – June, 2023**

PATHOPHYSIOLOGY-THEORY

Paper : BP204T

Time : Three hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Attempt ***all*** question from Section-A, ***two*** questions from Section-B and ***seven*** questions from Section-C.

SECTION – A

1. Answer the following in short :

2 × 10 = 20

(a) Define Pathogenesis.

(b) Define Stem cells.

P. T. O.

- (c) What is Gout ?
- (d) Define Mutation.
- (e) Define AIDS.
- (f) Define Carcinogens.
- (g) What is Leprosy ?
- (h) Define Etiology.
- (i) Define Heart failure.
- (j) Define cell injury.

SECTION – B

Note : Long answer type questions : Answer any *two* out of the following *three* questions : $10 \times 2 = 20$

2. What is Inflammation and explain its mechanism and pathophysiology ?

3. What is Cancer ? Describe its classification and pathophysiology in detail.
4. Describe mechanism and pathophysiology of Parkinson's.

SECTION – C

Note : Short Answer Questions : Answer any *seven* out of the following *nine* questions. $5 \times 7 = 35$

5. Describe types of Feedback system.
6. What is cell injury ? Explain in detail.
7. What is Hypertension ? Explain its pathophysiology.
8. Describe in detail about Diabetes and its pathophysiology.
9. What are the different disease of Joints ? Explain Rheumatoid arthritis in detail.

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10. Describe pathophysiology of Asthma.
 11. Enlist various disorders Sex hormones and explain its mechanism (any *two*).
 12. What are infectious diseases ? Describe pathophysiology of Typhoid in detail.
 13. What is Renal failure ? Explain in detail about acute and chronic Renal failure.
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Indira Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences
B. Pharm. 2nd Semester, 1st Sessional Examination, July, 2023
Paper Code –BP204T; Pathophysiology (Theory)

Duration: 1 hr 30 minutes

Total marks: 30

1. Question no. 1 is compulsory.

(5×2=10 marks)

- a. Define atrophy and hypertrophy with example.
- b. Define hyperplasia and metaplasia with example.
- c. Define acidosis and alkalosis with example.
- d. Define calcification with example.
- e. Define dysplasia with example.

2. Attempt any two questions from the rest.

(2×5=10 marks)

- a. Discuss the clinical symptoms and pathophysiology of asthma.
- b. Define homeostasis. Explain positive and negative feedback system.
- c. Write a brief note on the pathophysiology of atherosclerosis.

3. Attempt anyone questions from the rest.

(1×10=10 marks)

- a. Discuss the pathophysiology and the complications of Diabetes Mellitus in detail.
- b. Discuss the pathophysiology of hypertension in detail.

Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences
B. Pharm. 2nd Semester, 2nd Sessional Examination, August, 2023
Paper Code -BP204T; Pathophysiology (Theory)

Duration: 1 hr 30 minutes

Total marks: 30

1. Question no. 1 is compulsory.

(5×2=10 marks)

a. Pathogenesis of hypogonadism.

b. Pathogenesis of Polycystic Ovary Syndrome (PCOS).

c. Pathogenesis of Turner Syndrome

d. Pathogenesis of Gynecomastia.

e. Pathogenesis of hyperthyroidism.

2. Attempt any two questions from the rest.

(2×5=10 marks)

a. Discuss the clinical symptoms and pathophysiology of gout.

b. Write a brief note on the Alzheimer's disease

c. Write a brief note on the pathophysiology of epilepsy.

3. Attempt any one questions from the rest.

(1×10=10 marks)

a. Discuss the pathophysiological differences between of parkinson's disease and schizophrenia in detail.

b. Discuss the pathophysiology of peptic ulcer in detail.